

INDIAN STATISTICAL INSTITUTE
Mathematics-3 (BSDS)
QUIZ - 2 Examination

Date: 17/11/2025

Total Marks: 10

Time: 4:15 – 5:00 pm

**MARKS WILL BE AWARDED ONLY FOR THOSE ANSWERS WITH
PROPER JUSTIFICATION**

Question 1: [5 marks]

Let G be a connected bipartite graph with bipartite sets X and Y . Suppose that for any two adjacent vertices $x \in X$ and $y \in Y$, we have:

$$\deg(x) \geq \deg(y)$$

Prove that G has a matching that covers X .

Question 2: [5 marks]

Let (X, d) and (Y, d) be metric spaces. Define:

$$\bar{d}((x_1, y_1), (x_2, y_2)) := \max\{d(x_1, x_2), d(y_1, y_2)\} \quad ((x_i, y_i) \in X \times Y).$$

Show that $\bar{d}$ defines a metric on the product $X \times Y$.

– ALL THE BEST –